# Unstructured Data Analysis - Term Project Proposal

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## Topic: Automated Detection of Year-over-Year Changes in Corporate Annual Reports by Industry

- Compare annual reports of companies in the semiconductor, secondary battery, and biotechnology/pharmaceutical industries across multiple years
- Automatically detect major changes in business, products/services, financial information, and production facilities/investment
- Compare company-specific changes with common industry-level changes

## 1. Background & Motivation

### 1) Problem

- Annual reports contain key information on a company's business, products, R&D, and investment activities.
- Many sections are repeated across years, making it difficult and time-consuming to identify what has actually changed through manual comparison.
- Simple text or sentence-level comparison cannot effectively distinguish meaningful business changes from minor wording differences.

  → As a result, identifying important year-over-year changes and understanding their significance requires considerable time and effort.

### 2) Solution

- Utilize Natural Language Processing (NLP) to automate the identification of meaningful changes between annual reports from consecutive years.
- Focus on changes that reflect actual differences in a company's business and financial activities rather than simple textual variations.
- Extend the analysis from individual companies to industry-level patterns by examining changes across multiple companies within the same industry.

## 2. Project Objectives

- **Main Goal**: Automatically compare annual reports of the same company across different years.
- **Detect major changes in the following 4 areas:**
  - **Business Overview**: changes in business structure, industry/market environment, and major business activities
  - **Key Products and Services**: additions, discontinuations, or changes in the importance of major products and services
  - **Financial Information**: major changes in financial items and accounting-related issues, including potential risk signals

    - **Production Facilities and Investment**: changes in new facilities, production capacity, investment scale, and investment plans
- **Classify detected changes into the 5 following categories:**
    - New
    - Removed / Discontinued
    - Expanded: increases in scale, production capacity, investment, business scope, etc.
    - Reduced: decreases in scale, production capacity, investment, business scope, etc.
    - Modified: changes in business content, conditions, or structure that cannot be clearly classified into the four categories above
- Compare company-level changes with common industry-level changes by aggregating detected changes across companies within the same industry.
- Compare TF-IDF-based lexical similarity with Sentence Embedding-based semantic similarity to evaluate which method is more suitable for detecting meaningful year-over-year changes.

## 3. Solution & Data

### 1) Analysis Target

- Industries: Semiconductors / Secondary Batteries / Biotechnology & Pharmaceuticals
- 5 companies per industry, 15 companies in total
- Period: 2022–2025
- Up to 60 annual reports

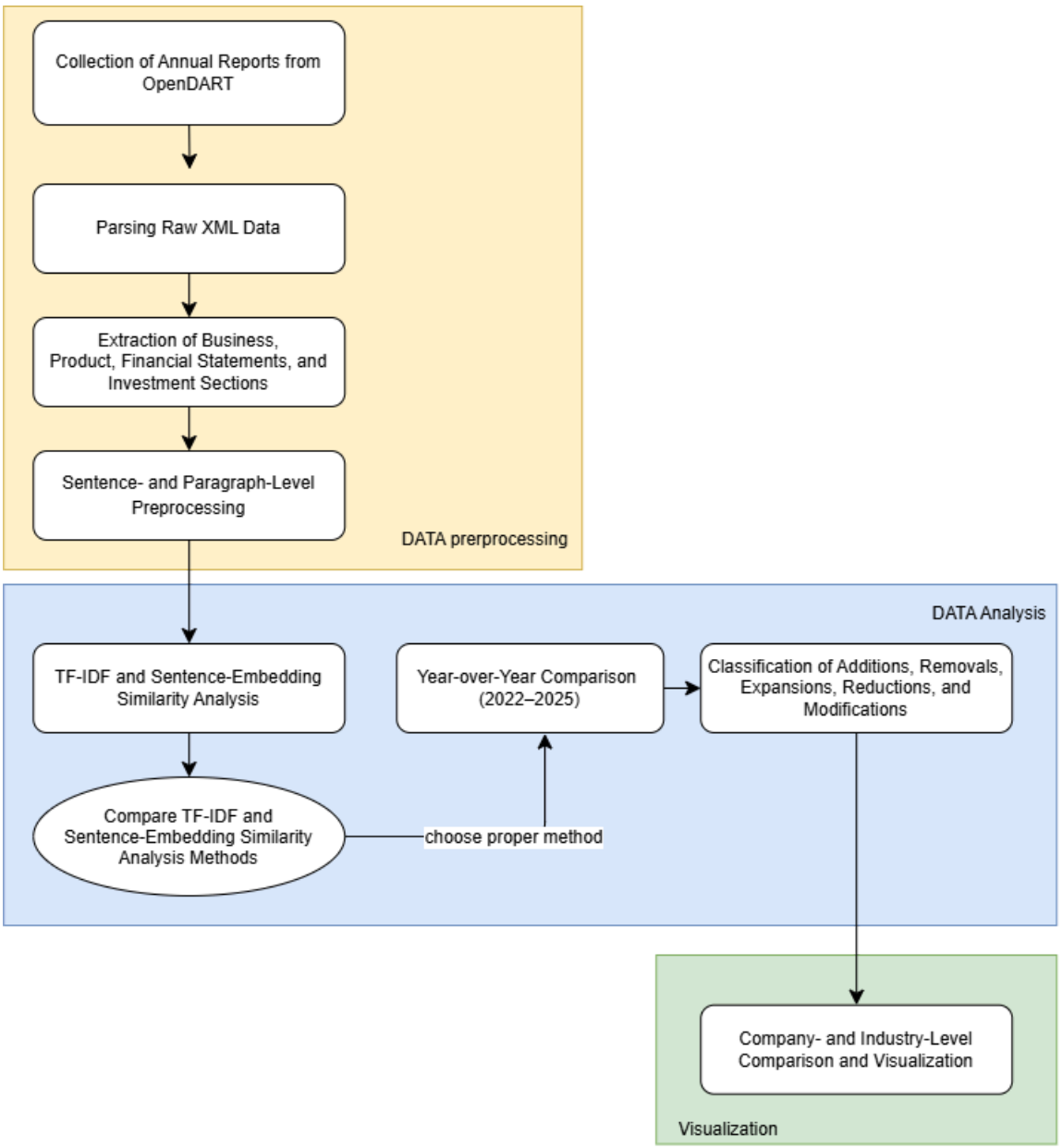
### 2) Reason for Selection

- These industries experience frequent changes in technology, products, financial conditions, investment, and market environments, which are reflected in annual reports.
- Their different business characteristics make them suitable for comparing year-over-year change patterns across industries.
- Analyzing multiple companies within each industry allows the project to identify both company-specific changes and common industry-level trends.

### 3) Data Sources

- 금융감독원 OpenDART
- Corporate Code API: identify each company's `corp_code`
- Disclosure Search API: collect the annual report `rcept_no`
- Original Disclosure Document API: download annual report source files in XML format

### 4) Analysis Pipeline




***comparison of two method***

- A subset of the data will be manually labeled as a test set.
- The labeled test set will be used as ground truth to evaluate the performance of the change detection methods.
- TF-IDF-based lexical similarity and Sentence Embedding-based semantic similarity will be compared using Precision, Recall, and F1-score.

## 4. Expected Impact

- Quickly identify key year-over-year changes in lengthy annual reports
- Distinguish simple wording changes from actual business changes
- Identify not only company-specific changes but also common patterns across industries

- Build an NLP-based analysis system using real-world unstructured disclosure data from DART
- Extend the system to detect changes in other types of corporate disclosure documents in the future

***Expected Outputs***

1. Company / Year / Section / Change Type + Before-and-After Evidence Sentences
2. Industry-level common change areas and change types for a selected period